PARALLEL REACTIVE CIRCUITS

AC CIRCUITS

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%	30%	**40%**
PRELIM	MIDTERM	ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Phasor Diagram

Parallel R-L Circuit

Parallel R-C Circuit

Parallel R-L-C Circuit

Power Factor

PHASOR DIAGRAM

R-X-Z COMPLEMENT

Ohms, Ω

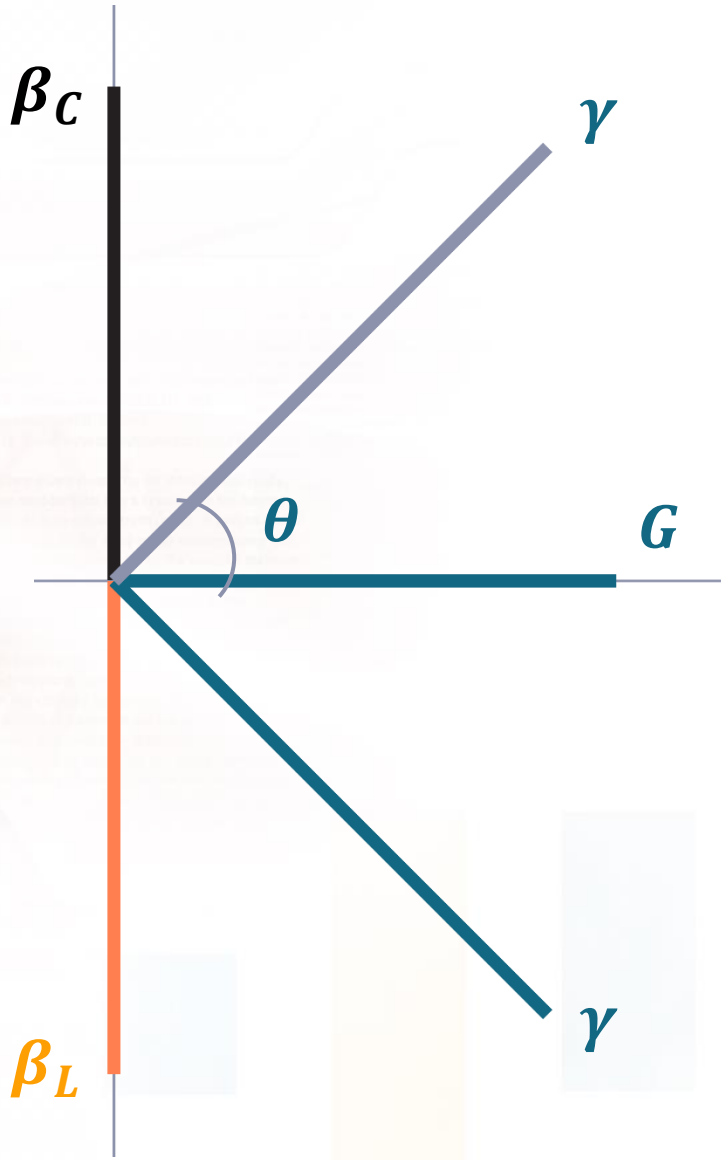
Resistance	Reactance	Impedance
R	X	Z

Siemens, S

Conductance	Susceptance	Admittance
$G = \frac{1}{R}$	$\beta = \frac{1}{X}$	$\gamma = \frac{1}{Z}$

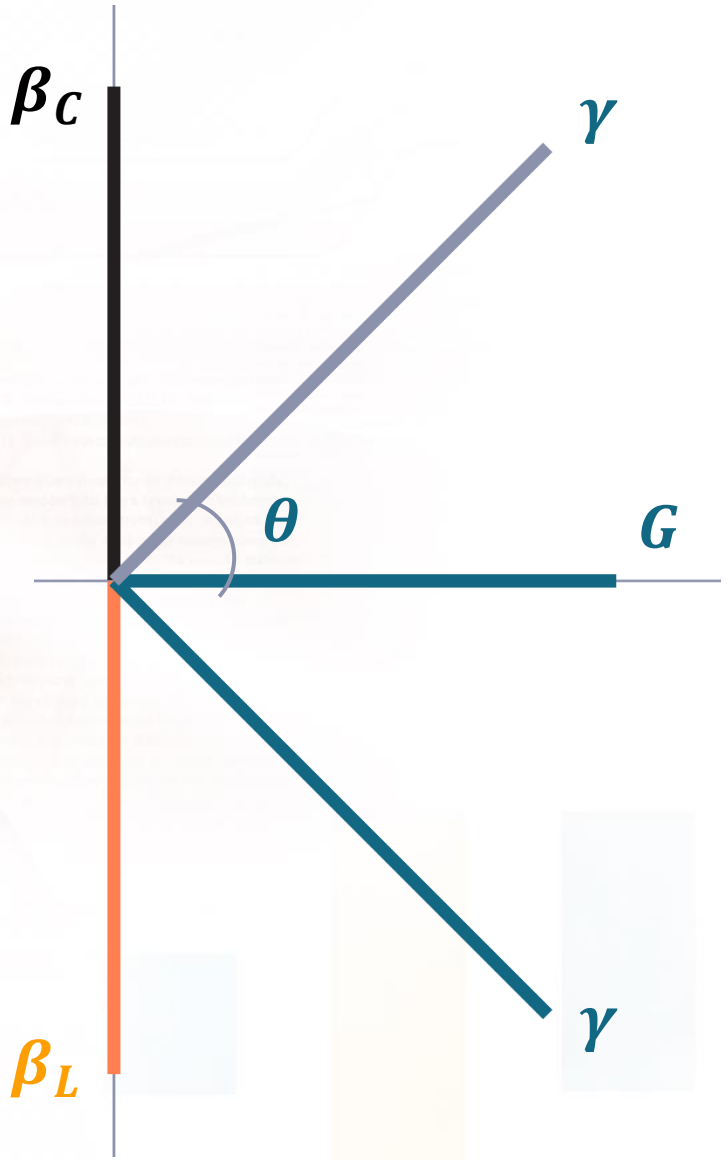
PHASOR DIAGRAM

Graphical representation of **magnitude** and **phase** relationship between sinusoidal quantities



The reciprocal of **impedance**,

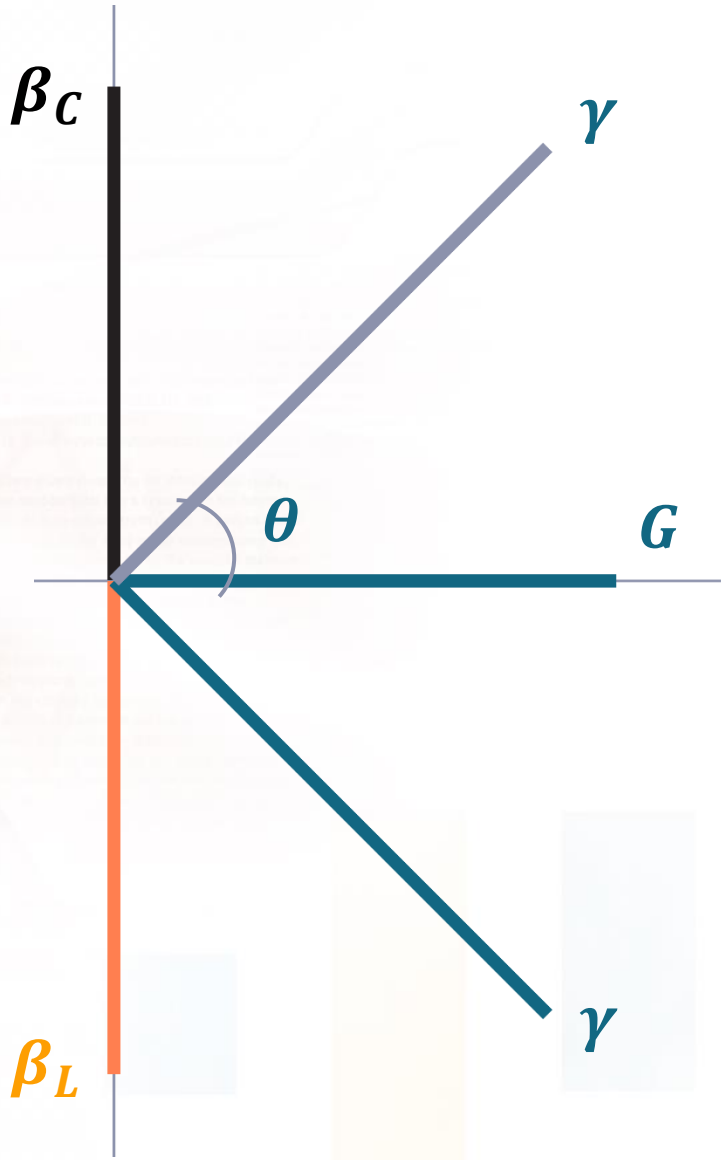
$$\gamma = \frac{1}{Z}$$



CAPACITIVE SUSCEPTANCE

The reciprocal of **capacitive reactance**,

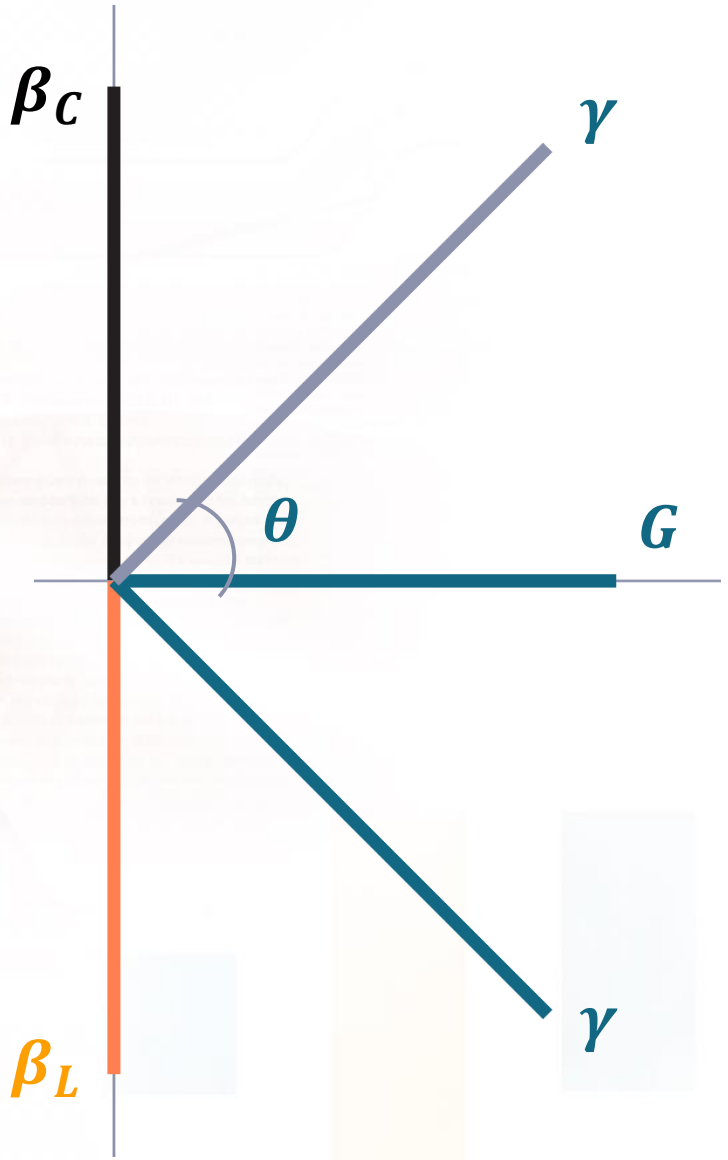
$$\beta_C = \frac{1}{X_C}$$



INDUCTIVE SUSCEPTANCE

The reciprocal of **inductive reactance**,

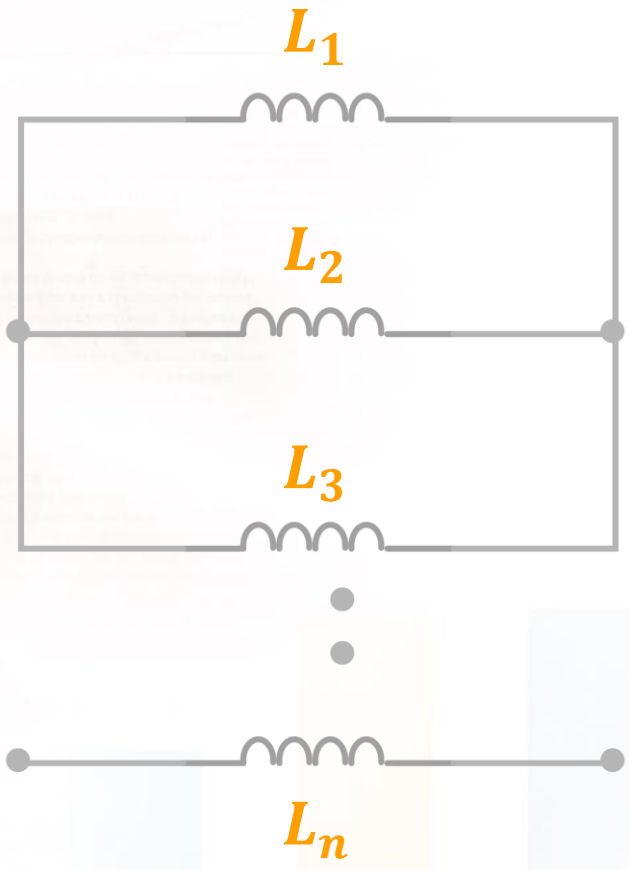
$$\beta_L = \frac{1}{X_L}$$



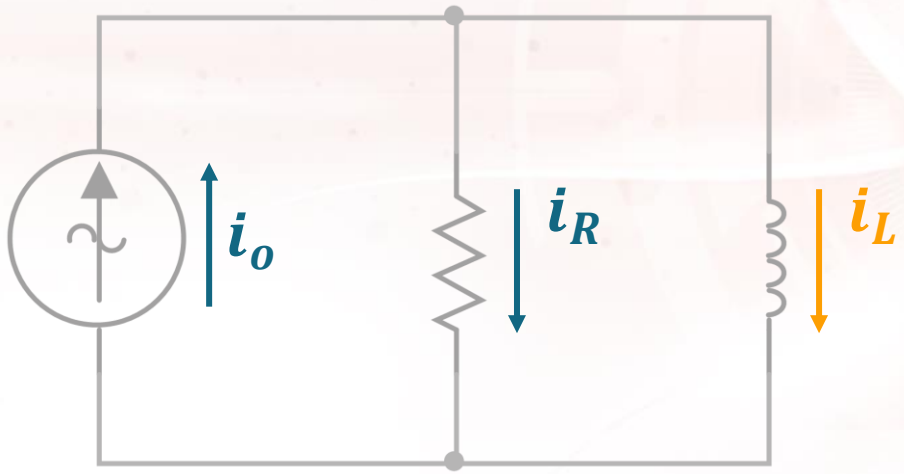
PARALLEL R-L CIRCUIT

Parallel,

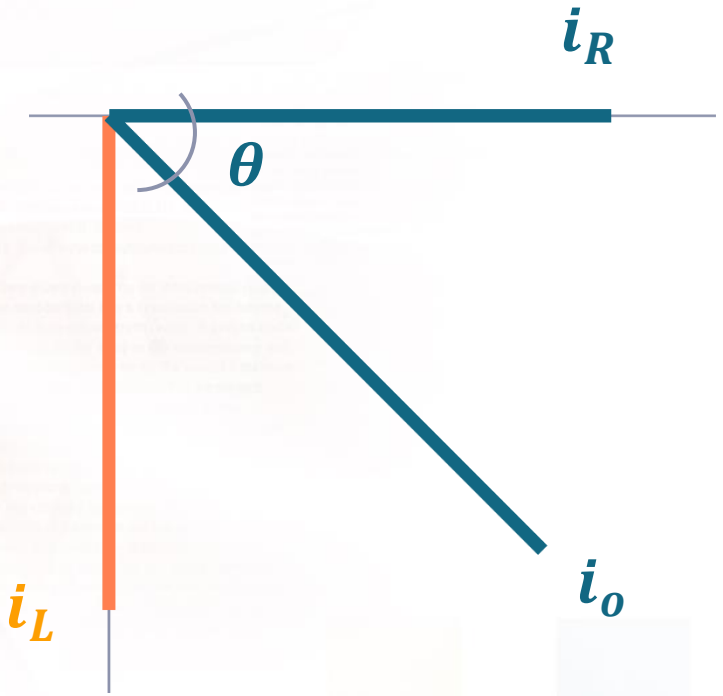
$$\frac{1}{L_o} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots \frac{1}{L_n}$$



$$i_o = i_R + j i_L$$

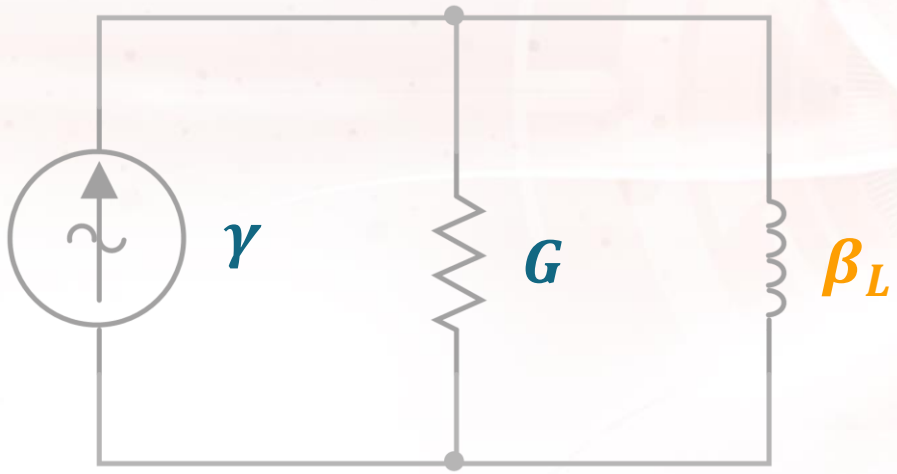


Circuit Diagram

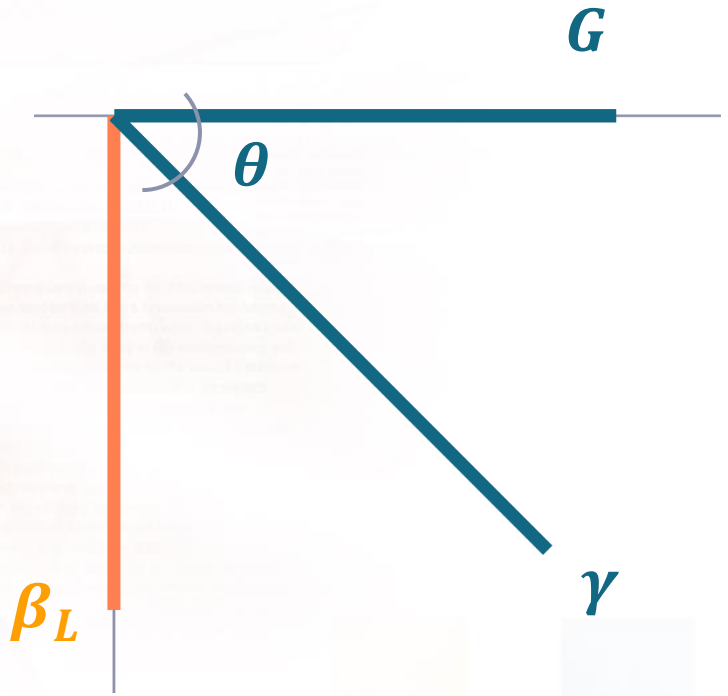


Current Phasor Diagram

$$\gamma = G + j\beta_L$$



Circuit Diagram



Admittance Phasor Diagram

A **120-volt 60-cycle** source is connected to a parallel combination of **30-ohm** resistor a **100-mH** inductor.

Calculate the following:

a) current through the resistor, i_R

b) current through the inductor, i_L

c) total circuit current, i_o

d) Tha phase angle between the voltage and current, θ

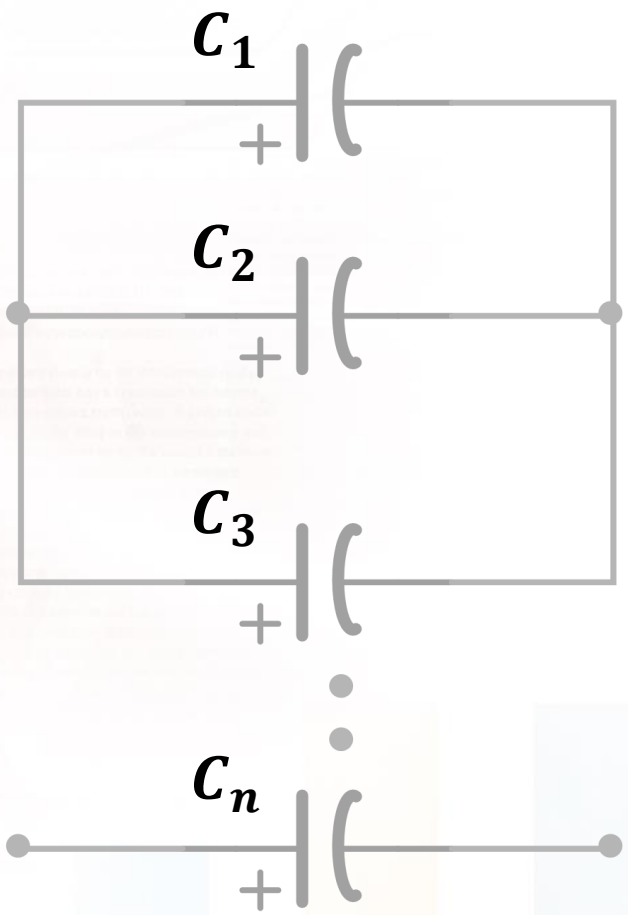
e) express all currents in **polar notation**

PARALLEL R-C CIRCUIT

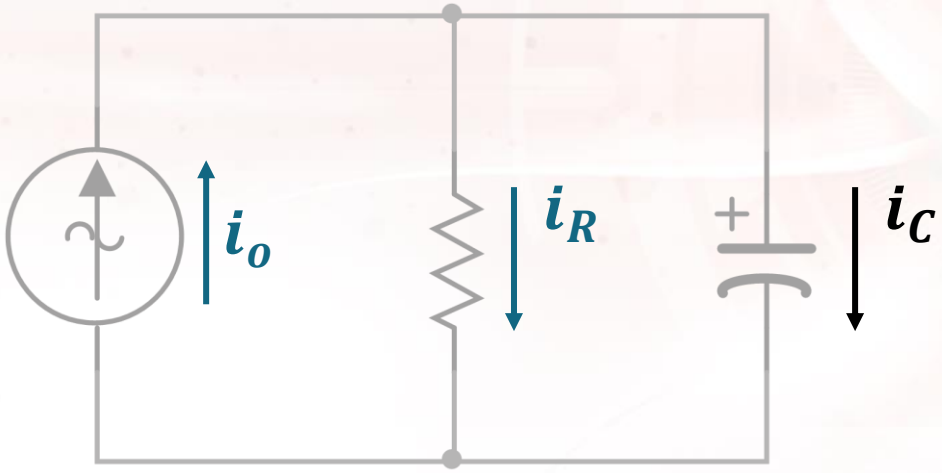
TOTAL CAPACITANCE

Parallel,

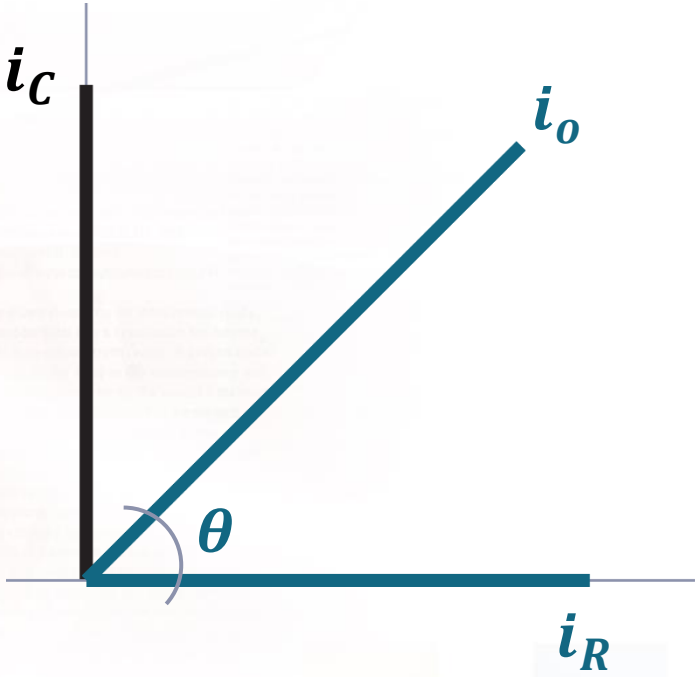
$$C_o = C_1 + C_2 + C_3 + \cdots C_n$$



$$i_o = i_R + j i_C$$

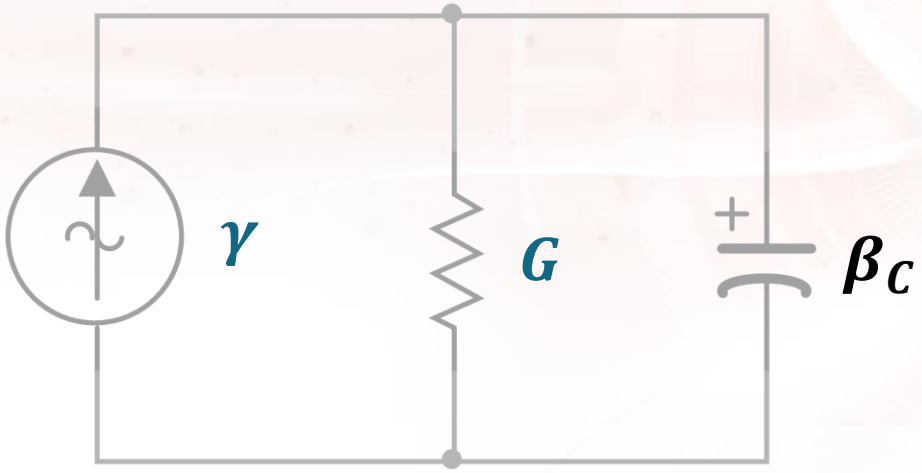


Circuit Diagram

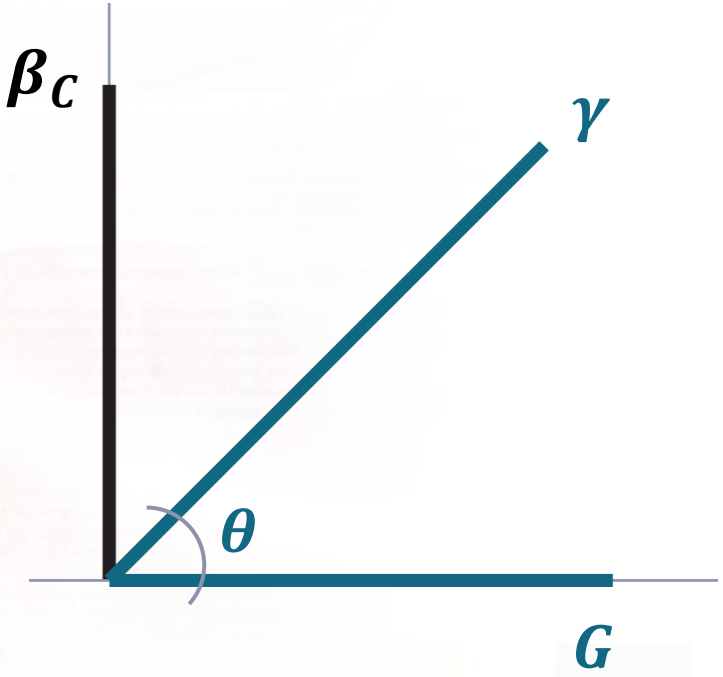


Current Phasor Diagram

$$\gamma = G + j\beta_C$$



Circuit Diagram



Admittance Phasor Diagram

A **120-volt 60-cycle** source is connected to a parallel combination of **10-ohm** resistor a **120-μf** capacitor.

Calculate the following:

a) current through the resistor, i_R

b) current through the capacitor, i_C

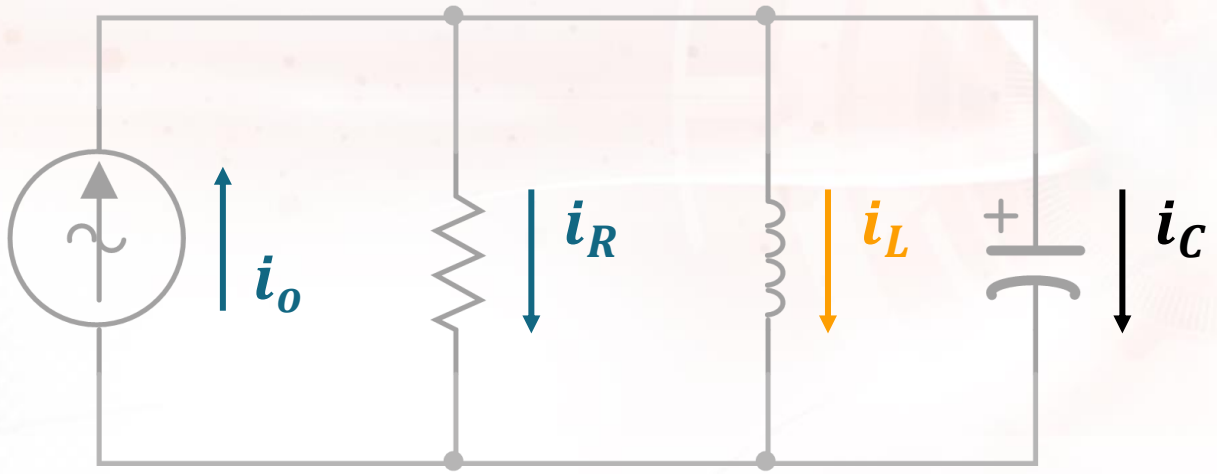
c) total circuit current, i_o

d) the phase angle between the voltage and current, θ

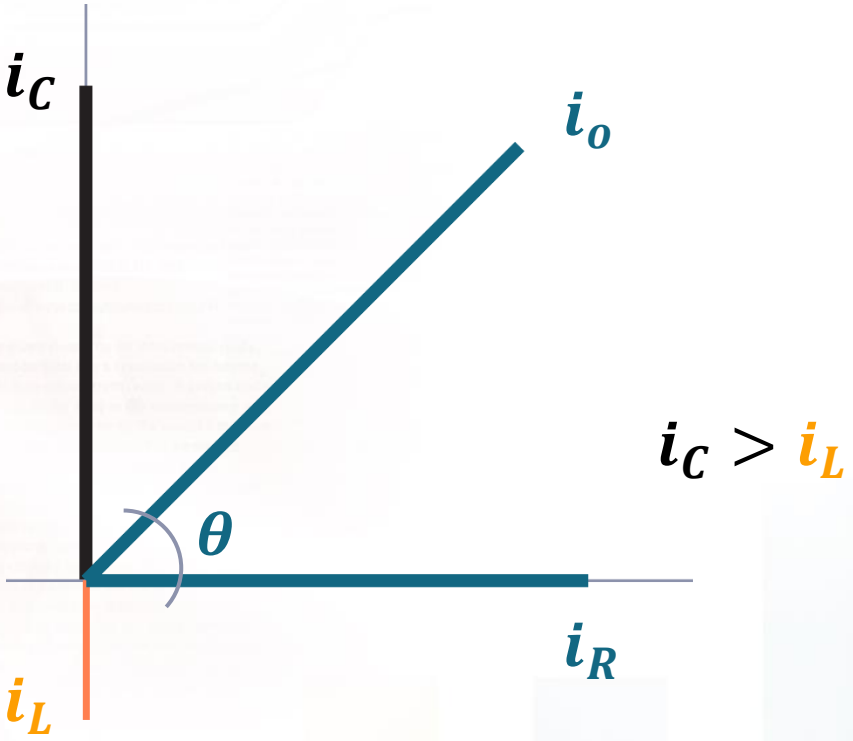
e) express all currents in **polar notation**

PARALLEL R-L-C CIRCUIT

$$i_o = i_R + j(i_C - i_L)$$

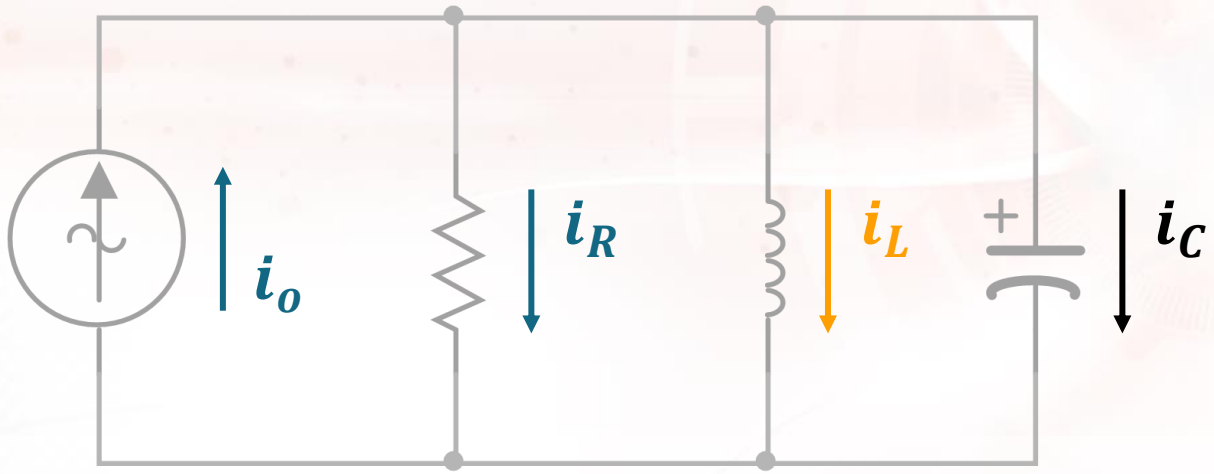


Circuit Diagram

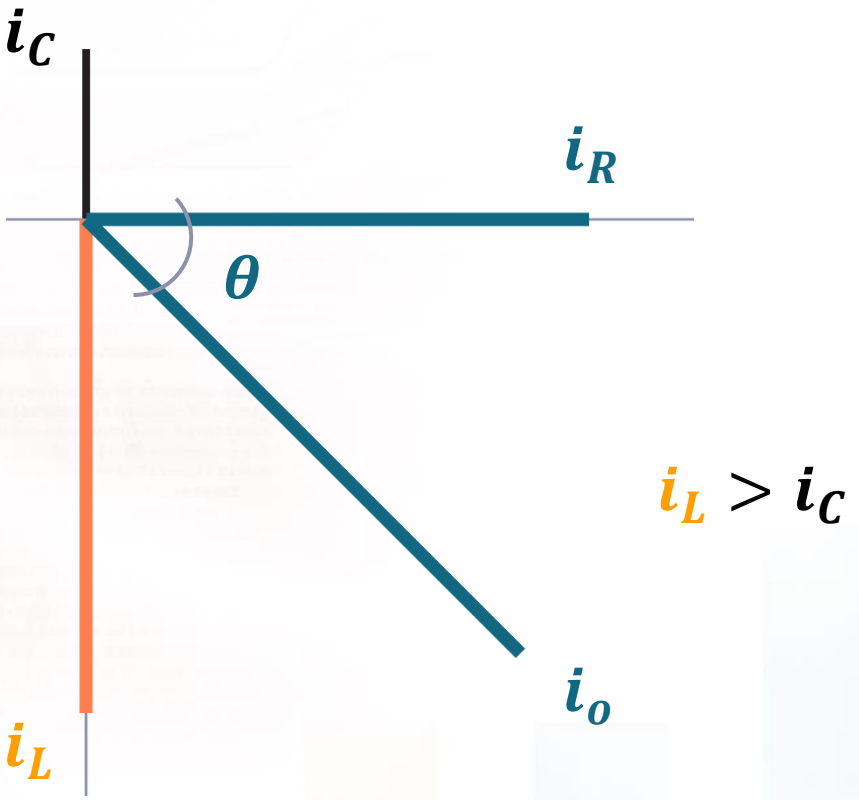


Current Phasor Diagram

$$i_o = i_R + j(i_C - i_L)$$



Circuit Diagram



Current Phasor Diagram

A **50-ohm** resistor, a **20-mH** coil and a **5-μf** capacitor are all connected in parallel across a **50-V, 100-Hz** supply.

Calculate the following:

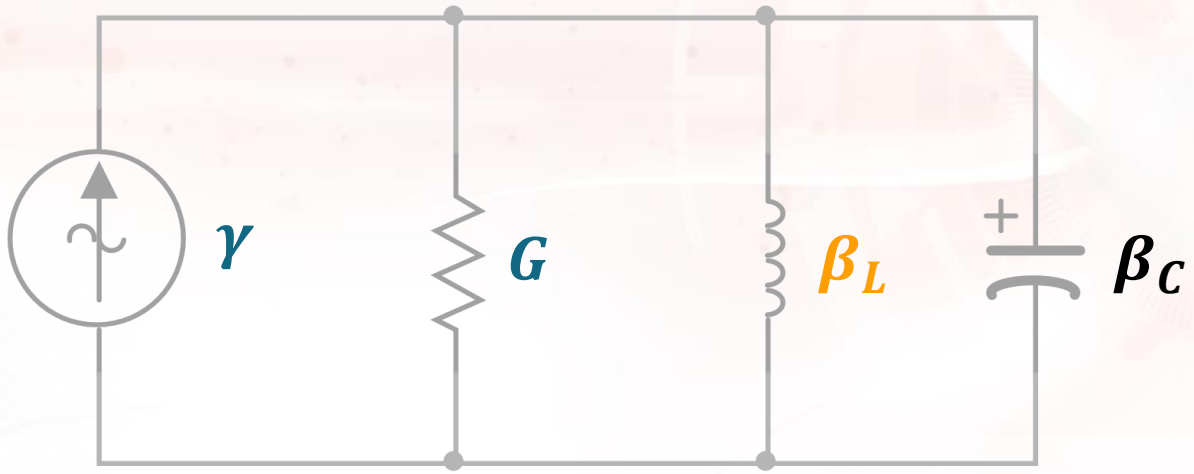
a) current through the resistor, i_R

b) current through the capacitor, i_C

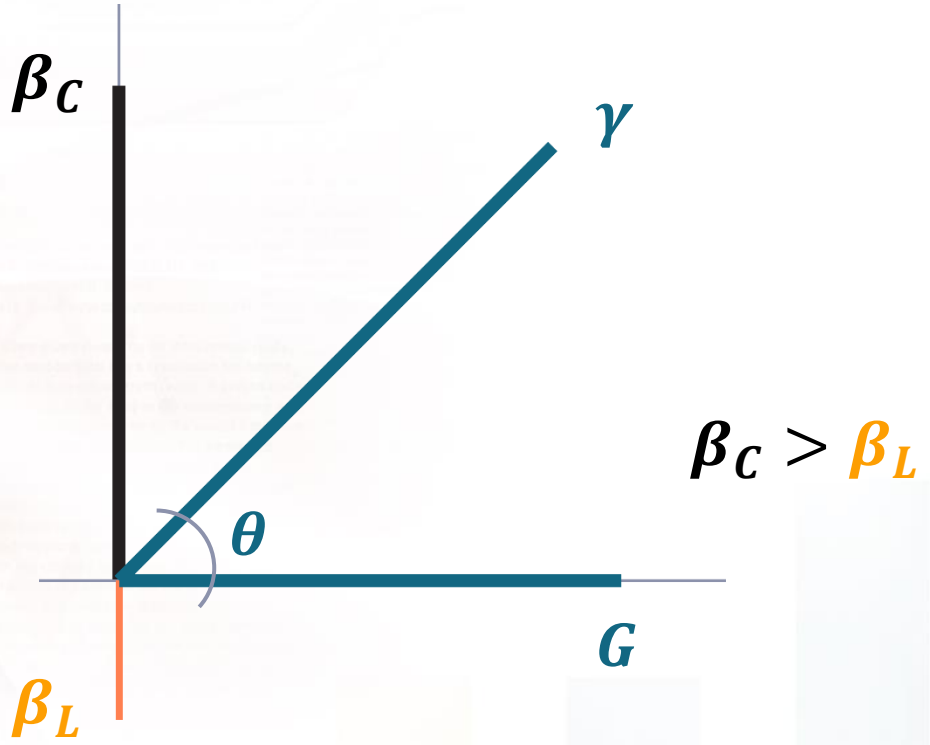
c) current through the inductor, i_L

d) total circuit current, i_o

$$\gamma = G + j(\beta_C - \beta_L)$$

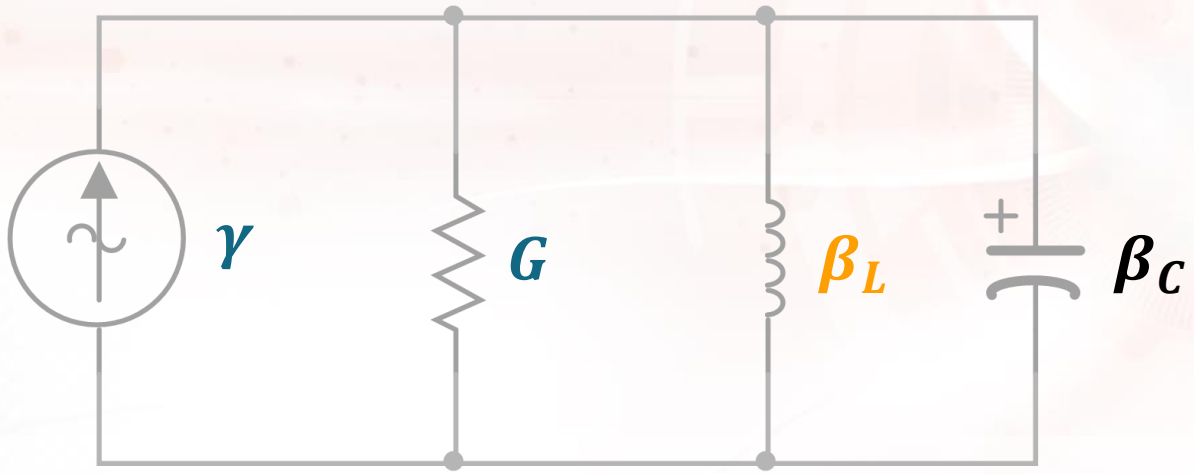


Circuit Diagram

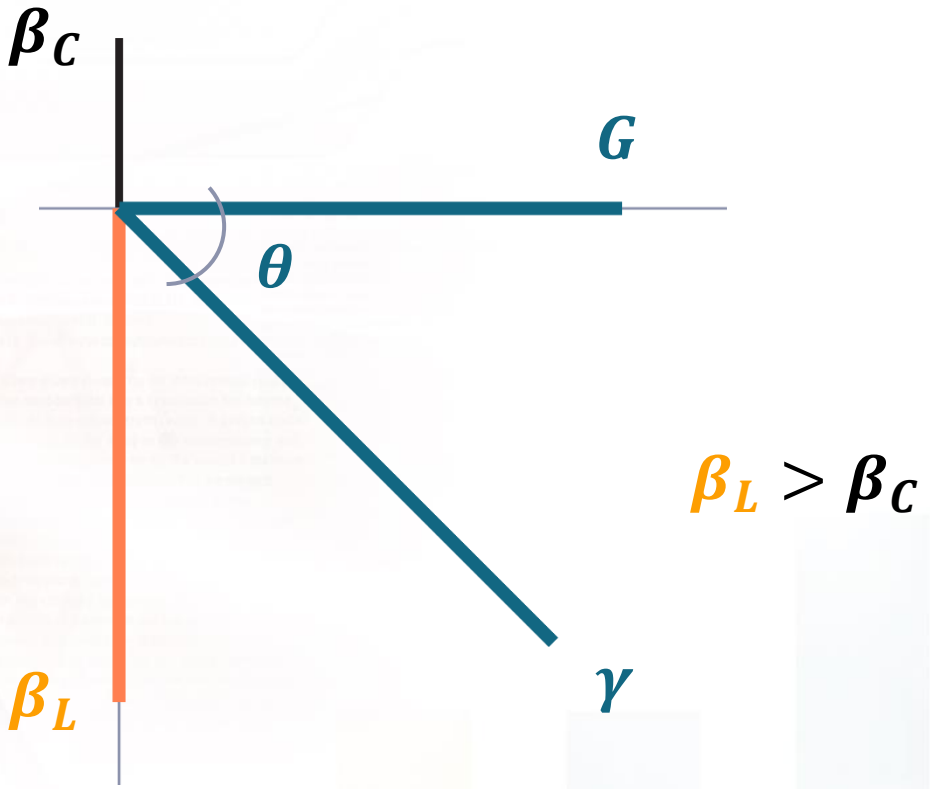


Admittance Phasor Diagram

$$\gamma = G + j(\beta_C - \beta_L)$$



Circuit Diagram



Admittance Phasor Diagram

A **50-ohm** resistor, a **20-mH** coil and a **5-μf** capacitor are all connected in parallel across a **50-V, 100-Hz** supply.

Calculate the following:

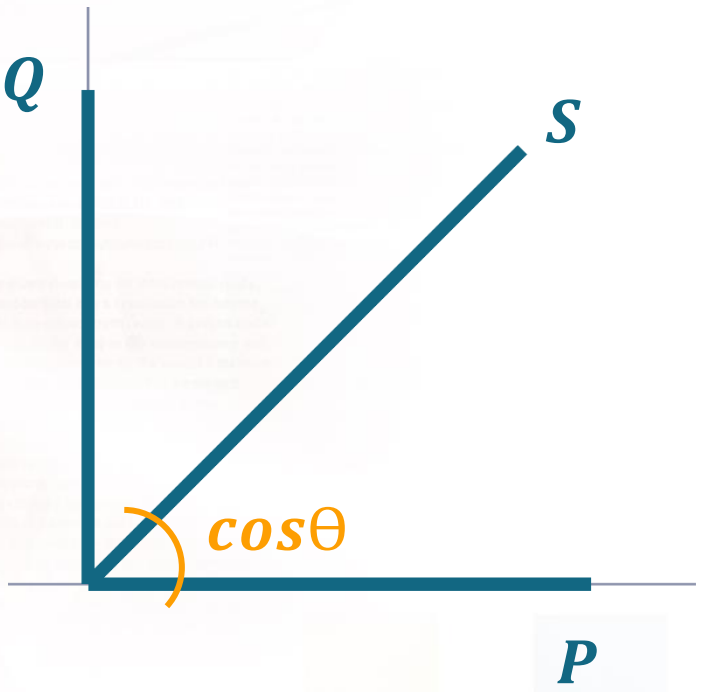
a) the total impedance of the circuit, Z

b) the phase angle between the voltage and current, θ

POWER FACTOR

The ratio of true power to apparent power

$$\cos\theta = \frac{P}{S}$$



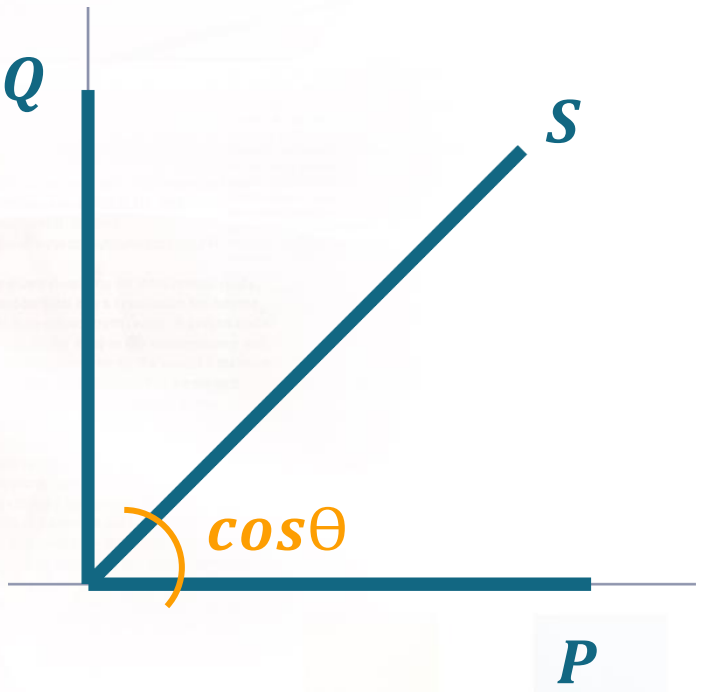
True Power, P

$$P = \frac{v^2}{R}$$

unit: watt (W)

The ratio of true power to apparent power

$$\cos\theta = \frac{P}{S}$$



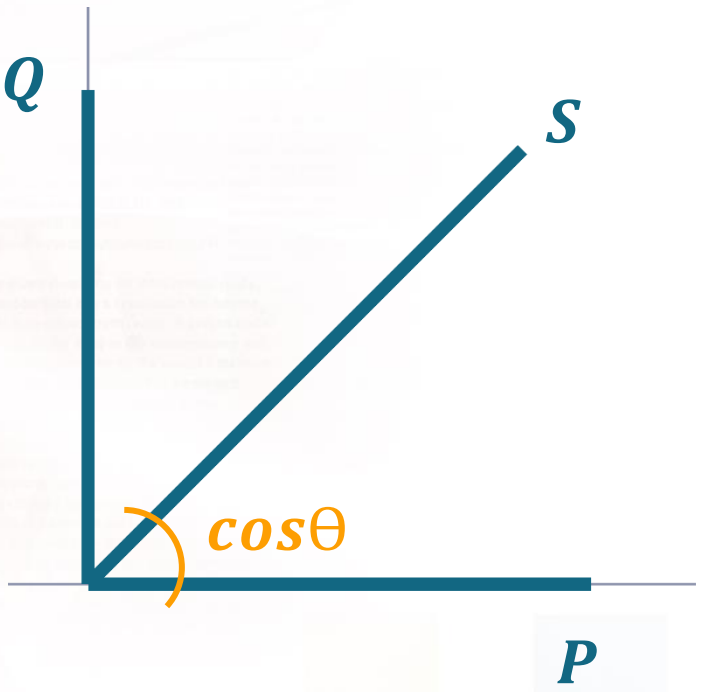
Reactive Power, Q

$$Q = \frac{v^2}{X}$$

unit: Volt-Amps-Reactive (VAR)

The ratio of true power to apparent power

$$\cos\theta = \frac{P}{S}$$



Reactive Power, Q

$$Q = \frac{v^2}{Z}$$

unit: Volt-Amps-Reactive (VAR)

A **50-ohm** resistor, a **20-mH** coil and a **5-μf** capacitor are all connected in parallel across a **50-V, 100-Hz** supply.

Calculate the following:

a) The **apparent, S** power and the **power factor, θ**.

LABORATORY